Amelia Burton

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Education

University of Illinois Urbana-Champaign May 2027
Bachelor of Science in Mechanical Engineering *GPA: 3.42/4.00*

Relevant Coursework: Mechanical Design, Design for Manufacturability, Fluid Dynamics, Heat Transfer, Micromanufacturing Processes, Welding & Metallurgy Processes

Engineering Experience

Woodward Inc. Rockford, IL
Senior Design Project *August 2026 – Present*

- Designed parametric CAD models of helical compression springs for aircraft fuel-control systems to study lateral forces and overturning moments.
- Applied design of experiments principles and Abaqus FEA to identify high-impact design parameters and compare spring geometries.
- Analyzed simulation results to identify designs that reduced lateral loading and improved valve performance.

Hensel Phelps Honolulu, HI
Construction Operations Intern *May 2026 – August 2026*

- Reviewed mechanical drawings, specifications, and submittals for piping, valves, pumps, ductwork, and equipment to identify design discrepancies.
- Authored 14 RFIs by reconciling contract documents with field conditions and coordinating multidisciplinary engineering issues.
- Performed field verification of equipment locations, dimensions, elevations, and installation conditions using Autodesk Navisworks for coordination.

University of California Observatories Santa Cruz, CA
Research and Development Intern *May 2024 – August 2024*

- Designed a mounting system for an approximately $80K CCD science camera, creating 9 custom components in SolidWorks.
- Produced GD&T manufacturing drawings and refined designs based on stakeholder requirements and fabrication constraints.
- Personally fabricated components using CNC/manual machining, forming, and welding and supported final hardware assembly.

Engineering Projects

Four-Speed Rover Transmission

- Designed the complete CAD model for a 4-speed transmission, including gears, shafts, and supporting components for a small rover.
- Built and tested the drivetrain, achieving 2 m/s, pulling a 3 kg load up a ramp, and surviving a 5 kg drop test; iterated the design based on physical testing.

Walking Dispenser

- Designed the complete CAD model for a Geneva-wheel and Scotch-yoke mechanism that dispensed one object every 2 m.
- Fabricated the mechanism using 3D printing and laser cutting and refined the design through testing to achieve 98% dispensing accuracy.

Technical Skills

Design & Analysis:	SolidWorks, Fusion 360, Onshape, GD&T, Abaqus, ANSYS, FEA, Design of Experiments
Programming & Tools:	Python, Excel, Navisworks, Revit, Bluebeam
Manufacturing:	CNC Mill, Manual Mill, Lathe, Welding, Waterjet, Sheet-Metal Forming, 3D Printing
Professional Development:	Planning to take the Mechanical Engineering FE Exam before graduation